

COMP 575 Sample MCQ

March 27, 2024

1 Multiple Choice Question

1. The difference between supervised and unsupervised learning is that :
 - (a) **Correct Answer :** The supervised learning allows the availability of a teacher to have some knowledge of the environment in terms of pattern labels, while the unsupervised learning only has patterns without desired responses.
 - (b) Supervised is used for single-layer or multi-layer neural networks, whereas unsupervised learning for recurrent networks.
 - (c) Supervised learning is only applicable to multi-layer neural networks, and unsupervised learning applicable to the self-organising maps, unless the data can undergo a data normalisation first to allow an unsaturated induced local field.
 - (d) Supervised learning is only possible for classification, whereas unsupervised for regression, regardless of the type of neural networks that can be used.
2. What is the main reason multi-layer perceptrons have multiple layers?
 - (a) **Correct Answer :** To learn nonlinearly separable problems.
 - (b) To accelerate convergence and learning.
 - (c) To enable the use of nonlinear activation functions, such as the sigmoid one.
 - (d) To allow easy calculation of the derivatives of the error with respect to the weights, when not all patterns are available simultaneously.
3. The quantity $\sum_{j=0}^p w_{kj}x_j$ (also called induced local field) of a neuron with p inputs x_j , multiplied by weights, w_{kj} assumes a term at $j = 0$. Why?
 - (a) **Correct Answer :** To accommodate a bias with a fixed input typically set to $x_0 = \pm 1$
 - (b) To accommodate a bias with a coupled input at $x_0 = x_p - \theta$, where θ is a function of the learning rate.

- (c) To allow recurrent connections to form, by connecting past inputs to current ones.
 - (d) To enable an optional memory element during training the neuron's weights.
4. The resulting optimal pattern x^* from the equation $x^* = \operatorname{argmin}_{x \in D_{train}} \|x - x_t\|_2$, where D_{train} is the training dataset, and x_t the unseen/query pattern, is used for which learning purpose?
- (a) **Correct Answer :** Used in memory-based learning to locate the stored pattern closest to the query pattern x_t , and assign it the known corresponding class/label of x^* .
 - (b) Used in multi-layer feed-forward networks to train the query pattern x_t , and find its output x^* .
 - (c) Used in competitive learning to locate how closely ordered the query pattern x_t is to the training dataset.
 - (d) Used in Boltzmann learning to derive the probability of flipping the state of a given neuron input x_t .
5. A fitness transformation scheme is of paramount importance to genetic algorithms because:
- (a) **Correct Answer :** It converts the objective values to fitnesses, so that the final selection probabilities can balance how individuals are selected for mating.
 - (b) Objective values do not really reflect the quality of the different individuals in the population, so transformed fitnesses can be disassociated.
 - (c) It converts the objective values to fitnesses, so that crossover and mutation become faster in searching the solution space.
 - (d) Objective values cannot really be used to quantify or predict the close-race or super-individual cases.
6. Enforcing simultaneously the elitism and the no-duplicates operators in genetic algorithms, means that during evolution:
- (a) **Correct Answer :** We cannot have any duplicate members in the population and also the best population objective can never decrease throughout evolution.
 - (b) We cannot have any duplicate members in the population apart from the elite individuals that have best objective or fitness values.
 - (c) We cannot have any elite members that will duplicate genetic material in their population as this is superfluous.

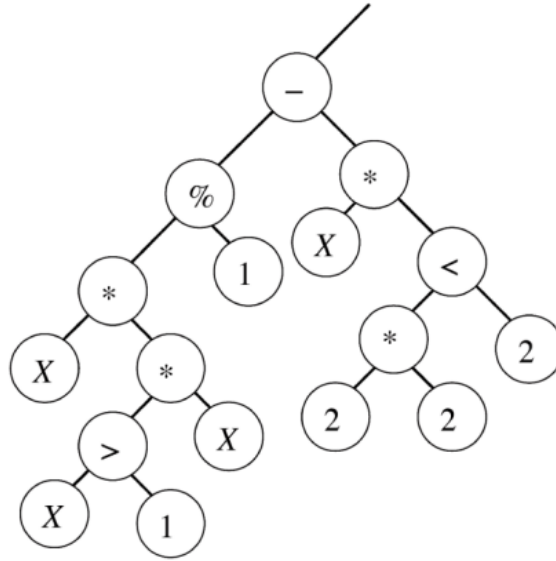


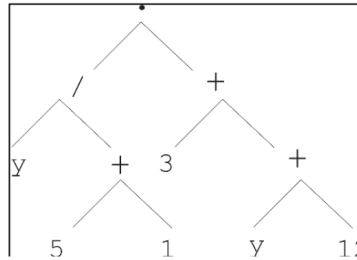
Figure 1: Figure for MCQ No. 8

- (d) We can only allow the steady-state model, which only produces very few individuals per generation, to guide evolution.
7. In the context of genetic programming, what is referred to as an intron and what are its implications?
- Correct Answer :** It is a subtree that bears no useful result in the phenotype, can waste a lot of computational resources but can also protect against destructive crossover operations.
 - It is a subtree that bears no useful result in the phenotype, but can pull the solution towards a global optimum in the solution space.
 - It is a subtree that has large function complexity, and it can protect against destructive crossover operations.
 - It is a subtree that has large depth, and it can protect against destructive mutation operations.
8. In the following genetic programming tree chromosome below, what is the complexity in terms of basic functions used, and which is the largest intron (if any) branch?
- Correct Answer :** The complexity is 8, and the intron is the expression " $X * (2 * 2 < 2)$ ", which always yields zero.
 - The complexity is 17, and the intron is the expression " $X * (2 * 2 < 2)$ ", which always yields zero.

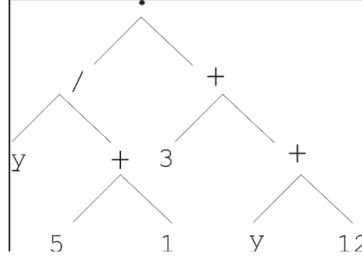
- (c) The complexity is 8, and the largest intron is the expression " $2 * 2 < 2$ ", which always yields zero.
 - (d) The complexity is 7, and the intron is the expression " $X > 1$ ", which is always false.
9. You are given a single-layer perceptron with two inputs x_1 and x_2 , and one threshold corresponding to a fixed input denoted as $x_0 = -1$. The perceptron is based on a signum activation function of output values +1 (for non-negative input) and -1 (for negative input). At some point, during the task of training, you observe your network to have weights: $w_0 = +1.5$ (the threshold) $w_1 = +2.5$, $w_2 = +0.4$ (weight indices correspond to input indices). Then, you need to train for pattern $(x_1, x_2) = (1, 0)$ with desired outcome $d = -1$, and adjust the weights accordingly, with a learning rate of 0.5. What will those weights become as soon as you train for this pattern?
- (a) **Correct Answer :** $(+2.5, +1.5, +0.4)$
 - (b) $(+2.5, +1.5, +1.4)$
 - (c) $(+2.5, +2.5, +0.4)$
 - (d) $(+3.5, +1.5, +0.4)$
10. Why does the linear-least-squares algorithm allow the minimisation of the error and therefore the recovery of the optimal weights in a single step, when all patterns are available simultaneously?
- (a) **Correct Answer :** Because direct application of the Gauss-Newton method to minimise the error, leads to a formula that depends on the Jacobian matrix which is equal to the negative of the data matrix; therefore the final weight-change formula does not depend on the current weights.
 - (b) Because direct application of the Gauss-Newton method to minimise the error, leads to a formula that depends on the Jacobian matrix, which does not need knowledge of all the patterns in the available training dataset.
 - (c) Because direct application of the optimisation leads to a linearly separable case where all patterns can be separated perfectly, so an iterative procedure is not necessary for the task of regression.
 - (d) The single step is permitted, as we do not have available multiple layers but the procedure is applicable to single layer neurons only.
11. Time-varying modifications in particle swarm optimisation are common and hence there are several variants. Most of them are based on common sense heuristics that can facilitate the search and the convergence. Which of the following does not make sense in terms of its associated rationale?

- (a) **Correct Answer :** The cognitive and social acceleration coefficients are adjusted so that the cognitive one increases during the iterations, while the social decreases, in order to allow the particles to wander through the entire space without clustering around a local optimum in the early generations but converge to a global optimum later.
 - (b) The inertia weight starts with a larger value to support more global search, while later on the weight decreases in order to favour local fine-tuning.
 - (c) The cognitive and social acceleration coefficients are adjusted so that the cognitive one decreases during the iterations, while the social increases, in order to allow the particles to wander through the entire space without clustering around a local optimum in the early generations but converge to a global optimum later.
 - (d) The social acceleration coefficient is adjusted so that it increases during the iterations, in order to allow the particles to converge to a global optimum later on towards the end of the search.
12. Which of the following descriptions is not correct about evolutionary strategies?
- (a) **Correct Answer :** They have been proven that they will always locate the global optimum in 1 out of 5 optimisation problems they are applied to.
 - (b) They are referred to as two-member population strategies, as the child competes with its parent, but multi-children multi-parents variations also exist.
 - (c) It can be mathematically proven, that asymptotically they will converge to the optimum, though this is not too useful in practice.
 - (d) They maintain one solution that changes to a new one, while encoding the step and direction of these jumps through variance and covariance information that can also adapted.
13. Which of the following statements related to genetic algorithms and discrete crossover is correct?
- (a) **Correct Answer :** For a high-point crossover, we have high distribution bias, high recombination power, and higher gene disruption.
 - (b) For a high-point crossover, we have high distribution bias, high recombination power, and lower gene disruption.
 - (c) For a high-point crossover, we have low distribution bias, high recombination power, and higher gene disruption.
 - (d) For a high-point crossover, we have high distribution bias, low recombination power, and higher gene disruption.

14. Assume the output $y_k(x) = \Phi(\sum_j w_{kj} \Phi(\sum_i w_{ji} x_i))$ of neuron k to also be the final output of the neural network. This quantity is shown here to depend on inputs x_i and weights w_{kj} and w_{ji} , and some activation function Φ . Which configuration does the above correspond to?
- (a) **Correct Answer :** A one-hidden-layer feed-forward network receives inputs x_i , then transforms them through hidden neurons indexed by j, and then reach an output neuron indexed by k.
 - (b) A two hidden layer feed-forward network receives inputs x_i , then transforms them through hidden neurons indexed by j, then proceed to a second hidden layer indexed by i, and then reach a output neuron indexed by k.
 - (c) A recurrent neural network combines current inputs x_i and past inputs y_j through the activation function to obtain the output indexed by k.
 - (d) A single layer perceptron is used to facilitate conversion from inputs to outputs twice and obtain a weighted average via $y_j = \Phi(\sum_i w_{ji} x_i)$.
15. Assume you have the genetic programming genotypic tree shown below and is subjected just once to a mutation operator you designed. Assume that this operator replaces randomly a single node anywhere in the tree (below the root) with a new randomly generated branch of arbitrary size and complexity. Which of the following expressions could not have been produced by this procedure and the given tree?



- (a) **Correct Answer :** $\frac{y}{5+y}(3 + \sqrt{y} + 12)$
 - (b) $\frac{y}{5+1}(3 + y + exp(12x))$
 - (c) $\frac{y}{5+1}(\sqrt{y}\sqrt{3} + y + 12)$
 - (d) $\frac{y}{5+(3+y+12)}(3 + y + 12)$
16. Assume you have the genetic programming genotypic tree shown below and is subjected just once to a mutation operator you designed. Assume that this operator replaces randomly a single node anywhere in the tree (below the root) with a new randomly generated branch of arbitrary size and complexity. Which of the following expressions could not have been produced by this procedure and the given tree?



- (a) **Correct Answer :** $\frac{y}{5+y}(3 + \sqrt{y} + 12)$
- (b) $\frac{y}{5+1}(3 + y + \exp(12x))$
- (c) $\frac{y}{5+1}(\sqrt{y\sqrt{3}} + y + 12)$
- (d) $\frac{y}{5+(3+y+12)}(3 + y + 12)$
17. You are asked to classify these 4 patterns, each with a feature vector denoted by x_i and associated output denoted by y_i , for $i = 1, 2, 3, 4$:

$$\begin{aligned} x_1 &= [+1, +1], & y_1 &= -1, \\ x_2 &= [-1, +1], & y_2 &= +1, \\ x_3 &= [+1, -1], & y_3 &= +1, \\ x_4 &= [-1, -1], & y_4 &= -1. \end{aligned}$$

Denoting the j -th element of the i -th pattern by $x_i(j)$, which of the following statements is the correct one?

- (a) **Correct Answer :** Transforming each i -th pattern to $[x_i(1)*x_i(2), x_i(1)]$, allows a perfect classification using linear support vector machines.
- (b) Transforming each i -th pattern to $[x_i(1) + x_i(2), x_i(1)]$, allows a perfect classification using linear support vector machines.
- (c) Transforming each i -th pattern to $[x_i(1) + x_i(2), x_i(2) - x_i(2)]$, allows a perfect classification using linear support vector machines.
- (d) Transforming each i -th pattern to $[x_i(1) + x_i(1) * x_i(2), x_i(2)]$, does not allow a perfect classification using linear support vector machines, but allows perfect classification using nonlinear kernel support vector machines.
18. Which of the following statements is not correct in the context of genetic algorithms and fitness utilisation?
- (a) **Correct Answer :** The close-race and super-individual cases are caused by very low and very high mutation rates, respectively.
- (b) The close-race effect is caused by low selective pressures, whereas the opposite leads to dominating individuals.

- (c) If the frequency of selecting fitter members is high then some super-individuals will dominate the population, whereas if it is low the search will become like a random walk.
- (d) Premature convergence can be caused when a small number of individuals contributing genetic material disproportionately to other individuals.

19. In gene expression programming, you are given a chromosome represented as:

$\text{+}\sqrt{\text{--}}/\text{b*abcd}$ abbbac

, where the head is the first 10 elements (enclosed by the oval), while the tail is the last 6 elements (enclosed by the box). Then, some mutation operator decides to replace the head's terminal symbol a with the maximum function (symbol M) that receives three arguments and returns the largest of the three. Which of the following corresponds to the correct resulting offspring with correctly demarcated (using the oval and the box shapes) head and tail?

$\text{+}\sqrt{\text{--}}/\text{b*Mbcdabb}$ bac

(a) **Correct Answer :**

$\text{+}\sqrt{\text{--}}/\text{b*Mbcd}$ bbbac

(b)

$\text{+}\sqrt{\text{--}}/\text{b*Mbcdabb}$ bac

(c)

$\text{+}\sqrt{\text{--}}/\text{b*Mbcd}$ abbbac

(d)

20. Why does the linear-least-squares (LLS) algorithm allow the minimisation of the error and therefore the recovery of the optimal weights in a single step, when all patterns are available simultaneously?

- (a) **Correct Answer :** Because direct application of the Gauss-Newton method to minimise the error, leads to a formula that depends on the Jacobian matrix which is equal to the negative of the data matrix, and as such, the final weight-finding formula does not depend on the current weights.
- (b) Because direct application of the Gauss-Newton method to minimise the error, leads to a formula that depends on the Jacobian matrix, which does not need knowledge of all the patterns in the available training dataset.
- (c) The single optimal weight-finding step is possible, as we do not have available multiple layers and therefore the procedure does not need to execute backward steps.

- (d) Because direct application of this particular optimisation procedure leads to a case where all patterns can be separated linearly, so an iterative procedure is not necessary for this particular task.